MODIFICATIONS OF A 2-DIMENSIONAL LOCAL DOMAIN ARE ISOMORPHISMS OVER THE PUNCTURED SPECTRUM

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Editorial note. Thin card at the H2 floor: the codimension observation (dim A = 2 forces closed points of U to be codimension 1) is genuine content beyond a bare pointer.

Allowed standard prerequisites (editorial).

- Modifications of normal schemes are isomorphisms in codimension 1. Spaces over Fields, lemma modification-normal-iso-over-codimension-1 (cross-chapter, named). Gives f^{-1}(U) -> U an isomorphism once every closed point of U has codimension 1.

Source reference. Original equation 0AE2: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 272–272. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BBQ: focused-seeds/geometry/S9-spres-modification/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a 2-dimensional Noetherian local domain such that $U = \mathrm{Spec}(A) \setminus \{\mathfrak{m}\}$ is a normal scheme. Then any modification $f : X \to \mathrm{Spec}(A)$ is a morphism as in (0AE2).*

Proof. Let $f : X \to S$ be a modification. We have to show that $f^{-1}(U) \to U$ is an isomorphism. Since every closed point u of U has codimension 1, this follows from Spaces over Fields, Lemma 0BBQ. □

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.